

Property Inheritance for Subtensors in Tensor Train Decompositions

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Abstract—Tensor dimensionality reduction is one of the fundamental tools for modern data science. To address the high computational overhead, fiber-wise sampled subtensors that preserve the original tensor rank are often used in designing efficient and scalable tensor dimensionality reduction. However, the theory of property inheritance for subtensors is still underdevelopment, that is, how the essential properties of the original tensor will be passed to its subtensors. This paper theoretically studies the property inheritance of the two key tensor properties, namely incoherence and condition number, under the tensor train setting. We also show how tensor train rank is preserved through fiber-wise sampling. The key parameters introduced in theorems are numerically evaluated under various settings. The results show that the properties of interest can be well preserved to the subtensors formed via fiber-wise sampling. Overall, this paper provides several handy analytic tools for developing efficient tensor analysis methods.

I. INTRODUCTION

The analytic study for tensor data, i.e., multi-dimensional array of numbers, has received much attention since last decade. Among many others, dimensionality reduction is one of the most popular methodologies for tensor data analysis, that is, to find a low-rank expression of the given tensor. Unlike the standard matrix rank definition, various ranks have been proposed for tensors; for instance, CP rank [1], Tucker rank [2], tubal rank [3], and tensor train rank [4]. The tensor dimensionality reductions have found a wide range of applications, such as signal processing [5]–[9], computer vision [10]–[14], social networks [15]–[17], and bioinformatics [18]–[21].

However, one major challenge for tensor dimensionality reduction is the high computational complexity associated with the complication of the high-dimensional structure. Recently, many Nystrom-style methods have been developed for tensor and matrix dimensionality reductions under various tensor rank settings [22]–[25]. These methods can significantly reduce the computational complexity of the dimensionality reductions via constructing and utilizing appropriated subtensors and submatrices which are potentially compact-sized. It is clear that good subtensors and submatrices are key to the success of Nystrom-style dimensionality reductions. While many studies have focused on the approximation theory aspect of the tensor Nystrom-style methods, such as sampling and perturbation analysis [24], [26], the theory of *property inheritance*, i.e.,

how the essential tensor properties such as incoherence and condition number, will be passed to subtensors, is still underdeveloped.

To address the vacancy in tensor theory, this paper studies the property inheritance for certain types of subtensors under the tensor train (TT) rank setting [4]. Following the common settings, we assume the original tensor is exactly low rank, and we focus on the subtensors that keep the original TT rank of the tensor. With the sequential nature of tensor train decomposition, we present results to explain how the ranks are sequentially determined in subtensors (see Theorem 2 and Corollary 3.) The subtensor property inheritance for incoherence and condition number is then presented in Theorems 4 and 5. In Section IV, we empirically evaluate the values of key parameters introduced in Theorems 4 and 5. The numerical results show that the tensor properties are well preserved with a simple sampling method. These results develop a deeper understanding of the properties of subtensors and can benefit future studies in related fields.

II. NOTATION AND PRELIMINARIES

We start with some basic notation. Distinct typefaces are used for different numerical structures. Specifically, calligraphic capital letters (e.g., $\mathcal{T}$) represent tensors, boldface capital letters (e.g., $\mathbf{M}$) denote matrices, regular capital letters (e.g., I) denote index sets, boldface lowercase letters (e.g., $\mathbf{v}$) are used for vectors, regular lower case letters (e.g., α) indicate scalars. The set of the first d natural numbers is denoted by $[d] := \{1, \dots, d\}$.

For a tensor $\mathcal{T}$, the notation $\mathcal{T}(I, :, \dots, :)$ refers to slicing or extracting a subset of the tensor where the indices in the first mode are restricted to I , while all indices in the other modes are selected. For a matrix $\mathbf{M} \in \mathbb{R}^{n_1 \times n_2}$, we use the following notations:

- $\mathbf{M}(I, :)$ denotes the $|I| \times n_2$ row submatrix of $\mathbf{M}$ consisting only of the rows indexed by $I \subseteq [n_1]$;
- $\mathbf{M}(:, J)$ denotes the $n_1 \times |J|$ column submatrix of $\mathbf{M}$ consisting only of the columns indexed by $J \subseteq [n_2]$;
- $\mathbf{M}(I, J)$ represents the $|I| \times |J|$ submatrix containing the entries a_{ij} of $\mathbf{M}$ for which $(i, j) \in I \times J$.

The $n \times n$ identity matrix is denoted as $\mathbb{I}_n$. We reserve the letters $\mathbf{W}_M$ and $\mathbf{V}_M$ to denote the left and right singular vectors of a matrix M . Finally, $M^\dagger$ denotes the Moore-Penrose pseudoinverse of M .

For the matrices, the two essential properties of interest, incoherence and condition number, are defined as follows.

Definition 1 (Matrix incoherence and condition number). *Let $M \in \mathbb{R}^{n_1 \times n_2}$ be a rank- r matrix, and let $M = \mathbf{W}_M \Sigma_M \mathbf{V}_M^\top$ be its compact SVD. Then M is said $\{\mu_{1,M}, \mu_{2,M}\}$ -incoherent (i.e., $\mu_{1,M}$ -column-incoherent and $\mu_{2,M}$ -row-incoherent) for some constants $\mu_{1,M}$ and $\mu_{2,M}$ such that*

$$\|\mathbf{W}_M\|_{2,\infty} \leq \sqrt{\frac{\mu_{1,M} r}{n_1}} \quad \text{and} \quad \|\mathbf{V}_M\|_{2,\infty} \leq \sqrt{\frac{\mu_{2,M} r}{n_2}}.$$

The condition number κ_M is defined as

$$\kappa_M := \frac{\sigma_{1,M}}{\sigma_{r,M}},$$

where $\sigma_{i,M}$ is the i -th largest singular value of M .

In the authors' prior work [27], property inheritance for submatrices has been thoroughly studied, i.e., how the incoherence and condition number of a given matrix will transfer to its row and column submatrices. The results can be summarized as the following theorem.

Theorem 1 (Section 3 of [27]). *Suppose $M \in \mathbb{R}^{n_1 \times n_2}$ is rank- r and $\{\mu_{1,M}, \mu_{2,M}\}$ -incoherent. Choose index set $I \subseteq [n_1]$ such that the row submatrix $\mathbf{R} = M(I, :)$ is also rank- r . Then it holds*

$$\begin{aligned} \mu_{2,\mathbf{R}} &\leq \mu_{2,M}, \\ \mu_{1,\mathbf{R}} &\leq \alpha^2 \kappa_M^2 \mu_{1,M}, \\ \kappa_{\mathbf{R}} &\leq \alpha \sqrt{\mu_{1,M} r} \kappa_M, \end{aligned}$$

where $\alpha := \sqrt{\frac{|I|}{n_1}} \|\mathbf{W}_M(I, :)^{\dagger}\|_2$. Similarly, choose index set $J \subseteq [n_2]$ such that the column submatrix $\mathbf{C} = M(:, J)$ is also rank- r . Then it holds

$$\begin{aligned} \mu_{1,\mathbf{C}} &\leq \mu_{1,M}, \\ \mu_{2,\mathbf{C}} &\leq \beta^2 \kappa_M^2 \mu_{2,M}, \\ \kappa_{\mathbf{C}} &\leq \beta \sqrt{\mu_{2,M} r} \kappa_M, \end{aligned}$$

where $\beta := \sqrt{\frac{|J|}{n_2}} \|\mathbf{V}_M(J, :)^{\dagger}\|_2$.

These results are handy tools for matrix analysis that involves submatrices. Naturally, researchers want to extend it to tensor settings. In fact, some recent work has successfully extended Theorem 1 to tensors under tubal setting [28], [29]. However, the study we proposed in this paper, i.e., extension to tensors in the TT setting, is rather complicated since the subtensors are obtained through sequential operations, just like TT decomposition itself.

Next, we introduce some basic preliminaries for tensor and tensor-train decomposition. For a thorough introduction, we refer the reader to [4], [30].

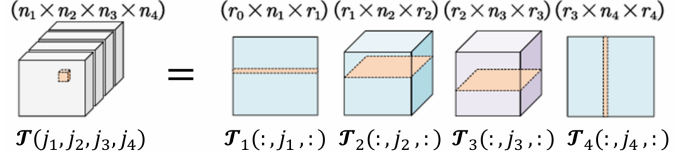


Fig. 1: [31]. Visual representation of tensor train (TT) decomposition for a 4-order tensor. Note that $r_0 = r_4 = 1$.

Definition 2 (Mode- k product). *Let $\mathcal{T} \in \mathbb{R}^{n_1 \times n_2 \times \dots \times n_d}$ and $M \in \mathbb{R}^{J \times n_k}$, the multiplication between $\mathcal{T}$ on its k -th mode with M is denoted as $\mathcal{X} = \mathcal{T} \times_k M$ with*

$$\begin{aligned} \mathcal{X}(i_1, \dots, i_{k-1}, j, i_{k+1}, \dots, i_d) \\ := \sum_{s=1}^{n_k} \mathcal{T}(i_1, \dots, i_{k-1}, s, i_{k+1}, \dots, i_d) M(j, s). \end{aligned}$$

Definition 3 (TT rank and i -th tensor unfolding). *Let $\mathcal{T} \in \mathbb{R}^{n_1 \times n_2 \times \dots \times n_d}$. The TT-rank of $\mathcal{T}$ is defined as $(r_1, r_2, \dots, r_{d-1})$, where $r_i := \text{rank}(\mathcal{T}_{\langle i \rangle})$, and the i -th tensor unfolding $\mathcal{T}_{\langle i \rangle} \in \mathbb{R}^{(n_1 \dots n_i) \times (n_{i+1} \dots n_d)}$ is obtained as:*

$$\mathcal{T}_{\langle i \rangle}(j_1 \dots j_i, j_{i+1} \dots j_d) = \mathcal{T}(j_1, j_2, \dots, j_d).$$

Note the i -th tensor unfolding defined here is different from the mode- i tensor unfolding.

Definition 4 (TT decomposition). *The TT decomposition of a tensor $\mathcal{T} \in \mathbb{R}^{n_1 \times n_2 \times \dots \times n_d}$ with TT-rank $(r_1, r_2, \dots, r_{d-1})$ is expressed as:*

$$\mathcal{T} := \mathcal{T}_1 \bullet \mathcal{T}_2 \bullet \dots \bullet \mathcal{T}_d,$$

where $\mathcal{T}_i \in \mathbb{R}^{r_{i-1} \times n_i \times r_i}$ are the core tensors, and $r_0 = r_d = 1$. Specifically, for $j_i \in [n_i]$, an entry of $\mathcal{T}$ can be written as:

$$\mathcal{T}(j_1, j_2, \dots, j_d) = \mathcal{T}_1(:, j_1, :) \mathcal{T}_2(:, j_2, :) \dots \mathcal{T}_d(:, j_d, :).$$

To help the reader better understand, Figure 1 is included here to illustrate TT decomposition.

Definition 5 (TT incoherence). *Let $\mathcal{T} \in \mathbb{R}^{n_1 \times n_2 \times \dots \times n_d}$ with TT-rank $(r_1, r_2, \dots, r_{d-1})$. Then $\mathcal{T}$ is said $\{\mu_{1,\mathcal{T}}, \mu_{2,\mathcal{T}}\}$ -incoherent, where $\mu_{1,\mathcal{T}}, \mu_{2,\mathcal{T}} \in \mathbb{R}^{d-1}$ and*

$$\mu_{1,\mathcal{T}}(i) := \mu_{1,\mathcal{T}_{\langle i \rangle}}, \quad \mu_{2,\mathcal{T}}(i) := \mu_{2,\mathcal{T}_{\langle i \rangle}}, \quad \forall i \in [d-1].$$

III. MAIN RESULTS

This section presents the main theoretical results for sub-tensor property inheritance under the tensor train (TT) setting.

Under the matrix setting, property inheritance is studied for row/column-wise sampled submatrices with the same rank as the original matrix [27]. The reason is simple: if a submatrix has the rank of the original matrix, then it spans the same linear subspace, thus preserving the subspace information of the original matrix. Similarly, this paper aims at the subtensors with the same TT rank as the original tensor. As shown in Definition 4, TT decomposition is computed as a series of operations on tensor dimensions in a sequential order. This contrasts with some other tensor decompositions, such as CP and Tucker, whose operations are in no particular order

of dimensions. Hence, to study the subtensors of interest, we must understand how fiber-wise sampling in the earlier dimension may impact the rank in later dimensions in TT decomposition.

As a base case, this study finds that if we fiber-wise samples along the first dimension to form a subtensor such that that rank of the mode-1 unfolding, i.e., r_1 , is preserved, then this subtensor will also preserve the rest of TT rank. This result is presented as Theorem 2.

Theorem 2. Let $\mathcal{T} \in \mathbb{R}^{n_1 \times n_2 \times \dots \times n_d}$ with $\text{TT-rank}(\mathcal{T}) = (r_1, r_2, \dots, r_{d-1})$. Suppose that the index set $I \subseteq [n_1]$ is chosen such that $\text{rank}(\mathcal{T}_{(1)}(I, :)) = r_1$. Then $\text{TT-rank}(\mathcal{T}(I, :, \dots, :)) = (r_1, r_2, \dots, r_{d-1})$.

Proof. Set subtensor $\mathcal{R} = \mathcal{T}(I, :, \dots, :)$. Since $\text{rank}(\mathcal{T}_{(1)}(I, :)) = r_1$, i.e., $\text{rank}(\mathcal{R}_{(1)}) = r_1$, by the theory of tensor CUR decomposition [24, Theorem 2,3], there exists $J \subseteq [n_2 n_3 \dots n_d]$ such that

$$\mathcal{T} = \mathcal{R} \times_1 \mathbf{C} \mathbf{U}^\dagger,$$

where $\mathbf{C} = \mathcal{T}_{(1)}(:, J)$ and $\mathbf{U} = \mathcal{T}_{(1)}(I, J)$. Thus, we have:

$$\mathcal{T}_{(i)} = (\mathbf{C} \mathbf{U}^\dagger \otimes \mathbb{I}_{d_2 \dots d_i}) \mathcal{R}_{(i)}. \quad (1)$$

From (1), it follows that

$$\text{rank}(\mathcal{T}_{(i)}) \leq \text{rank}(\mathcal{R}_{(i)}) = r_i. \quad (2)$$

Since $\mathcal{R}_{(i)}$ is a submatrix of $\mathcal{T}_{(i)}$, we also have

$$\text{rank}(\mathcal{T}_{(i)}) \geq \text{rank}(\mathcal{R}_{(i)}) = r_i. \quad (3)$$

Combining (2) and (3), we conclude that

$$\text{rank}(\mathcal{T}_{(i)}) = \text{rank}(\mathcal{R}_{(i)}) = r_i.$$

for all $i \in [d-1]$. This finishes the proof. $\square$

Naturally, we can extend this result to the subtensors formed by fiber-wise sampling along the i -th tensor unfolding, i.e., sample along the first dimension of the i -th tensor unfolding of the original tensor, and preserve the subsequent part of TT rank $\{r_j\}_{j=i}^{d-1}$. This result is presented as Corollary 3.

Corollary 3. Let $\mathcal{T} = \mathcal{T}_1 \bullet \mathcal{T}_2 \bullet \dots \bullet \mathcal{T}_d \in \mathbb{R}^{n_1 \times n_2 \times \dots \times n_d}$ with $\text{TT-rank}(\mathcal{T}) = (r_1, r_2, \dots, r_{d-1})$. Given a fixed $i \in [d-1]$, suppose that the index set $I \subseteq \bigotimes_{j=1}^i [n_j]$ is chosen such that $\text{rank}(\mathcal{T}_{(i)}(I, :)) = r_i$. Set $\mathcal{R} = (\mathcal{T}_1 \bullet \dots \bullet \mathcal{T}_i)_{(i)}(I, :) \bullet \mathcal{T}_{i+1} \bullet \dots \bullet \mathcal{T}_d$. Then $\text{TT-rank}(\mathcal{R}) = (r_i, r_{i+1}, \dots, r_{d-1})$.

Proof. This result is a direct combination of the definition of TT decomposition and Theorem 2. $\square$

Note that Corollary 3 does not account for potential fiber-wise sampling that occurs in the dimensions preceding the i -th dimension. Due to the sequential nature of TT decomposition, preserving all TT ranks in the formed subtensors, namely $\mathcal{R}_i$, requires that the index pool of fibers available for sampling in the later dimensions is constrained by the indices sampled in the preceding dimensions. Specifically,

$$\begin{cases} I_0 = \{1\}, \\ I_i \subseteq I_{i-1} \otimes [n_i] \subseteq \bigotimes_{j=1}^i [n_j] \end{cases} \quad (4)$$

for all $i \in [d-1]$, where I_i denotes the index set sampled in i -th tensor unfolding.

In Theorem 4, we present the property inheritance, in terms of TT incoherence and condition number, for the subtensors $\mathcal{R}_i$ formed with index sets I_i , given that the subtensors preserve the original TT rank.

Theorem 4. Let $\mathcal{T} = \mathcal{T}_1 \bullet \mathcal{T}_2 \bullet \dots \bullet \mathcal{T}_d \in \mathbb{R}^{n_1 \times n_2 \times \dots \times n_d}$ with $\text{TT-rank}(\mathcal{T}) = (r_1, r_2, \dots, r_{d-1})$ and $\{\mu_{1,\mathcal{T}}, \mu_{2,\mathcal{T}}\}$ -incoherence. With index sets I_i defined as (4), set

$$\mathcal{R}_i = (\mathcal{T}_1 \bullet \dots \bullet \mathcal{T}_i)_{(i)}(I_i, :) \bullet \mathcal{T}_{i+1} \bullet \dots \bullet \mathcal{T}_d$$

for $i \in [d-1]$. Note that $\mathcal{R}_i \in \mathbb{R}^{|I_i| \times n_{i+1} \times \dots \times n_d}$. Suppose that I_i are chosen such that $\text{rank}((\mathcal{R}_i)_{(1)}) = r_i$ for all i . Then $\{\mu_{1,\mathcal{R}}, \mu_{2,\mathcal{R}}\}$ satisfies the following conditions: for all $i \in [d-1]$ and $t \in [d-i]$,

$$\begin{aligned} \mu_{1,(\mathcal{R}_i)_{(t)}} &\leq \alpha_{i,t}^2 \kappa_{\mathcal{T}_{(t+i-1)}}^2 \mu_{1,\mathcal{T}_{(t+i-1)}}, \\ \mu_{2,(\mathcal{R}_i)_{(t)}} &\leq \mu_{2,\mathcal{T}_{(t+i-1)}}, \\ \kappa_{(\mathcal{R}_i)_{(t)}} &\leq \alpha_{i,t} \sqrt{\mu_{1,\mathcal{T}_{(t+i-1)}} r_{t+i-1}} \kappa_{\mathcal{T}_{(t+i-1)}}, \end{aligned}$$

where

$$\alpha_{i,t} := \sqrt{\frac{|I_i|}{\prod_{j=1}^i n_j}} \left\| \mathbf{W}_{\mathcal{T}_{(t+i-1)}} \left(I_i \otimes \left(\bigotimes_{j=i+1}^{t+i-1} [n_j] \right), : \right)^\dagger \right\|_2. \quad (5)$$

Proof. Firstly, let's fix a $i \in [d-1]$. Since the index set I_i gives $\text{rank}((\mathcal{R}_i)_{(1)}) = r_i$, by Corollary 3, we have that $\text{rank}((\mathcal{R}_i)_{(t)}) = \text{rank}(\mathcal{T}_{(t+i-1)})$ for $t \in [d-i]$. We observed that $(\mathcal{R}_i)_{(t)}$ for a fixed $t \in [d-i]$ is a row submatrix of $\mathcal{T}_{(t+i-1)}$ with row index $I_i \otimes \left(\bigotimes_{j=i+1}^{t+i-1} [n_j] \right)$. Applying Theorem 1 and set

$$\alpha_{i,t} = \sqrt{\frac{|I_i|}{\prod_{j=1}^i n_j}} \left\| \mathbf{W}_{\mathcal{T}_{(t+i-1)}} \left(I_i \otimes \left(\bigotimes_{j=i+1}^{t+i-1} [n_j] \right), : \right)^\dagger \right\|_2,$$

we have that

$$\begin{aligned} \mu_{1,(\mathcal{R}_i)_{(t)}} &\leq \alpha_{i,t}^2 \kappa_{\mathcal{T}_{(t+i-1)}}^2 \mu_{1,\mathcal{T}_{(t+i-1)}}, \\ \mu_{2,(\mathcal{R}_i)_{(t)}} &\leq \mu_{2,\mathcal{T}_{(t+i-1)}}, \\ \kappa_{(\mathcal{R}_i)_{(t)}} &\leq \alpha_{i,t} \sqrt{\mu_{1,\mathcal{T}_{(t+i-1)}} r_{t+i-1}} \kappa_{\mathcal{T}_{(t+i-1)}}. \end{aligned}$$

The above argument applies to any $i \in [d-1]$ and $t \in [d-i]$. Thus, it finishes the proof. $\square$

Those subtensors can be viewed as generalized row submatrices of the unfoldings. Since the ‘‘columns’’ are original, we see no amplification on the ‘‘column’’ incoherence parameter μ_2 while the ‘‘row’’ incoherence μ_1 can be slightly amplified due to fiber-wise sampling. The condition number can also become slightly worse through the sampling.

Now that the ‘‘row’’ samplings along the first to $(d-1)$ -st dimensions have been handled, we shift the focus to the properties of subtensors related to the ‘‘column’’ samplings. Similar to the matrix case where rows and columns can be

sampled independently, the indices of generalized column samplings on the unfoldings, namely J_i , can be independent of I_i . Specifically,

$$J_i \subseteq \left[\prod_{j=i+1}^d n_j \right] \quad (6)$$

for i -th tensor unfolding. The property inheritance for the subtensors formed by both I_i and J_i , namely $\mathbf{C}_i = (\mathcal{R}_{i-1})_{\langle 1 \rangle}(:, J_i)$, is presented as Theorem 5.

Theorem 5. Let $\mathcal{T} = \mathcal{T}_1 \bullet \mathcal{T}_2 \bullet \cdots \bullet \mathcal{T}_d \in \mathbb{R}^{n_1 \times n_2 \times \cdots \times n_d}$ with $\text{TT-rank}(\mathcal{T}) = (r_1, r_2, \dots, r_{d-1})$ and $\{\mu_{1,\tau}, \mu_{2,\tau}\}$ -incoherence. With index sets I_i and J_i defined as (4) and (6) respectively, set

$$\begin{cases} \mathcal{R}_0 = \mathcal{T}, \\ \mathcal{R}_i = (\mathcal{T}_1 \bullet \cdots \bullet \mathcal{T}_i)_{\langle i \rangle}(I_i, :) \bullet \mathcal{T}_{i+1} \bullet \cdots \bullet \mathcal{T}_d, \\ \mathbf{C}_i = (\mathcal{R}_{i-1})_{\langle 1 \rangle}(:, J_i) \end{cases}$$

for $i \in [d-1]$. Suppose that I_i and J_i is chosen such that $\text{rank}((\mathcal{R}_i)_{\langle 1 \rangle}) = r_i$ and $\text{rank}(\mathbf{C}_i) = r_i$ for all i . Then it holds

$$\begin{aligned} \mu_{1,\mathbf{C}_1} &\leq \mu_{1,\mathcal{T}_{(1)}}, \\ \mu_{2,\mathbf{C}_1} &\leq \beta_1^2 \kappa_{\mathcal{T}_{(1)}}^2 \mu_{2,\mathcal{T}_{(1)}}, \\ \kappa_{\mathbf{C}_1} &\leq \beta_1 \sqrt{\mu_{2,\mathcal{T}_{(1)}} r_1 \kappa_{\mathcal{T}_{(1)}}} \end{aligned}$$

for $i = 1$, and

$$\begin{aligned} \mu_{1,\mathbf{C}_i} &\leq \alpha_i^2 \beta_i^2 \kappa_{\mathcal{T}_{(i)}}^2 r_i \mu_{1,\mathcal{T}_{(i)}} \mu_{2,\mathcal{T}_{(i)}}, \\ \mu_{2,\mathbf{C}_i} &\leq \beta_i^2 \kappa_{\mathcal{T}_{(i)}}^2 \mu_{2,\mathcal{T}_{(i)}}, \\ \kappa_{\mathbf{C}_i} &\leq \alpha_i \beta_i \sqrt{\mu_{1,\mathcal{T}_{(i)}} \mu_{2,\mathcal{T}_{(i)}} r_i \kappa_{\mathcal{T}_{(i)}}}, \end{aligned}$$

for $2 \leq i \leq d-1$, where

$$\begin{aligned} \alpha_i &:= \sqrt{\frac{|I_{i-1}|}{\prod_{j=1}^{i-1} n_j}} \|\mathbf{W}_{\mathcal{T}_{(i)}}(I_{i-1} \otimes [n_i], :)^{\dagger}\|_2, \\ \beta_i &:= \sqrt{\frac{|J_i|}{\prod_{j=i+1}^d n_j}} \|\mathbf{V}_{\mathcal{T}_{(i)}}(J_i, :)^{\dagger}\|_2. \end{aligned} \quad (7)$$

Note that α_1 is not used in the theorem; however, we can take $\alpha_1 = 1$ since $\mu_{1,\mathbf{C}_1} \leq \mu_{1,\mathcal{T}_{(1)}}$ with an implicit α_1 .

Proof. Firstly, let's consider the case of $i = 1$. That is, $\mathbf{C}_1 = \mathcal{T}_{(1)}(:, J_1)$. Since J_1 is chose such that $\text{rank}(\mathbf{C}_1) = r_1$, by applying Theorem 1, we have that

$$\begin{aligned} \mu_{1,\mathbf{C}_1} &\leq \mu_{1,\mathcal{T}_{(1)}}, \\ \mu_{2,\mathbf{C}_1} &\leq \beta_1^2 \kappa_{\mathcal{T}_{(1)}}^2 \mu_{2,\mathcal{T}_{(1)}}, \\ \kappa_{\mathbf{C}_1} &\leq \beta_1 \sqrt{\mu_{2,\mathcal{T}_{(1)}} r_1 \kappa_{\mathcal{T}_{(1)}}}, \end{aligned}$$

where

$$\beta_1 := \sqrt{\frac{|J_1|}{\prod_{j=2}^d n_j}} \|\mathbf{V}_{\mathcal{T}_{(1)}}(J_1, :)^{\dagger}\|_2.$$

Next, we consider the cases of $2 \leq i \leq d-1$. Notice that $\mathbf{C}_i = (\mathcal{R}_{i-1})_{\langle 1 \rangle}(:, J_i) = \mathcal{T}_{(i)}(I_{i-1} \otimes [n_i], J_i)$. Additionally,

we are given $\text{rank}(\mathbf{C}_i) = r_i$ with the chosen J_i . By applying Theorem 1, we have that

$$\begin{aligned} \mu_{1,\mathcal{T}_{(i)}}(:, J_i) &\leq \mu_{1,\mathcal{T}_{(i)}}, \\ \mu_{2,\mathcal{T}_{(i)}}(:, J_i) &\leq \beta_i^2 \kappa_{\mathcal{T}_{(i)}}^2 \mu_{2,\mathcal{T}_{(i)}}, \\ \kappa_{\mathcal{T}_{(i)}}(:, J_i) &\leq \beta_i \sqrt{\mu_{2,\mathcal{T}_{(i)}} r_i \kappa_{\mathcal{T}_{(i)}}}, \end{aligned}$$

where $\beta_i = \sqrt{\frac{|J_i|}{\prod_{j=i+1}^d n_j}} \|\mathbf{V}_{\mathcal{T}_{(i)}}(J_i, :)^{\dagger}\|_2$.

Invoking Theorem 1 again, we have that

$$\begin{aligned} \mu_{1,\mathbf{C}_i} &\leq \alpha_i^2 \kappa_{\mathcal{T}_{(i)}}^2 (\mu_{1,\mathcal{T}_{(i)}}(:, J_i)) \mu_{1,\mathcal{T}_{(i)}}(:, J_i) \\ &\leq \alpha_i^2 \beta_i^2 \kappa_{\mathcal{T}_{(i)}}^2 r_i \mu_{1,\mathcal{T}_{(i)}} \mu_{2,\mathcal{T}_{(i)}}, \\ \mu_{2,\mathbf{C}_i} &\leq \mu_{2,\mathcal{T}_{(i)}}(:, J_i) \\ &\leq \beta_i^2 \kappa_{\mathcal{T}_{(i)}}^2 \mu_{2,\mathcal{T}_{(i)}}, \end{aligned}$$

and

$$\begin{aligned} \kappa_{\mathbf{C}_i} &\leq \alpha_i \sqrt{\mu_{1,\mathcal{T}_{(i)}}(:, J_i) r_i \kappa_{\mathcal{T}_{(i)}}(:, J_i)} \\ &\leq \alpha_i \beta_i \sqrt{\mu_{1,\mathcal{T}_{(i)}} \mu_{2,\mathcal{T}_{(i)}} r_i \kappa_{\mathcal{T}_{(i)}}}, \end{aligned}$$

where $\alpha_i = \sqrt{\frac{|I_{i-1}|}{\prod_{j=1}^{i-1} n_j}} \|\mathbf{W}_{\mathcal{T}_{(i)}}(I_{i-1} \otimes [n_i], :)^{\dagger}\|_2$. This completes the proof. $\square$

Remark 6. The bounds of property inheritance presented in Theorems 4 and 5 rely on the parameters $\alpha_{i,t}$, α_i , and β_i as defined in (5) and (7), which involves the Moore-Penrose pseudoinverses of subsampled left and right singular vectors of the unfoldings. The spectral bounds of these pseudoinverses highly depend on how the index sets I_i and J_i are sampled, and some sampling strategies may improve those parameters. This has been thoroughly discussed for matrix setting in [27]. However, due to page limits, we will only empirically verify those parameters (see Section IV) and reserve the theoretical discussion for future work.

IV. NUMERICAL EVALUATION

The values of the parameters $\alpha_{i,t}$, α_i , and β_i , as introduced in Theorems 4 and 5, are key to controlling the essential properties of the subtensors. As discussed in Remark 6, they rely on the sampling method for the index sets. This section numerically evaluates these parameters with one of the most simple yet popular methods—*uniform sampling without replacement*. All experiments are implemented on Matlab R2022b and executed on a laptop equipped with Intel i7-11800H CPU and 16GB DDR4 RAM.

For test data, we generate four-dimensional tensors $\mathcal{T} \in \mathbb{R}^{100 \times 100 \times 100 \times 100}$ with $\text{TT-rank}(\mathcal{T}) = (r_1, r_2, r_3) = (2, 3, 2)$ using the following three random methods:

- **Gaussian generation:** Set $\mathcal{T} = \mathcal{T}_1 \bullet \mathcal{T}_2 \bullet \mathcal{T}_3 \bullet \mathcal{T}_4$, where each entry of $\mathcal{T}_i \in \mathbb{R}^{r_{i-1} \times 100 \times r_i}$ is independently sampled from a Gaussian distribution with mean 0 and variance 1.
- **Hadamard generation:** Set $\mathcal{T} = \mathcal{T}_1 \bullet \mathcal{T}_2 \bullet \mathcal{T}_3 \bullet \mathcal{T}_4$, where each entry of $\mathcal{T}_i \in \{-1, 1\}^{r_{i-1} \times 100 \times r_i}$ is sampled

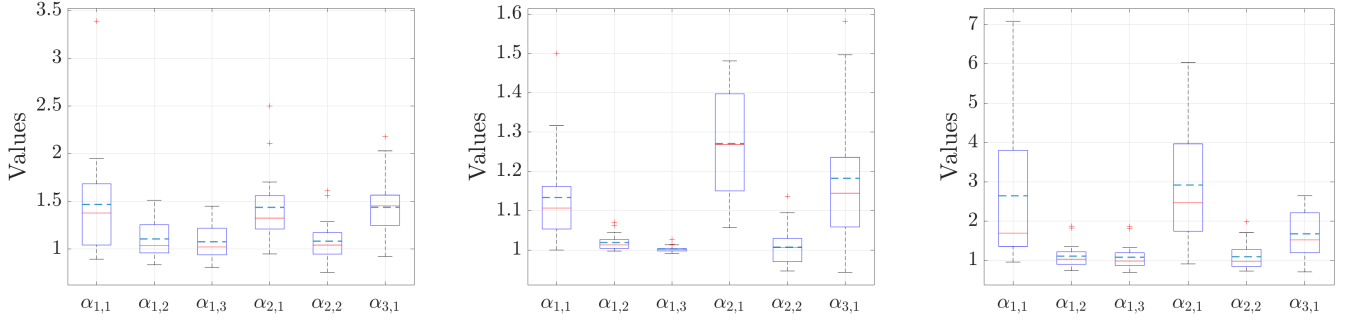


Fig. 2: Boxplot for $\alpha_{i,t}$ as introduced in Theorem 4. Each box represents the distribution of parameter values over 20 trials, showing the median (center line), interquartile range (box), and potential outliers (red +). The whiskers (top and bottom horizontal lines) extend to the most extreme data points within 1.5 times the interquartile range. The dashed blue line indicates the mean of the parameter values. **Left:** Gaussian generation; **Middle:** Hadamard generation; **Right:** Uniform generation.

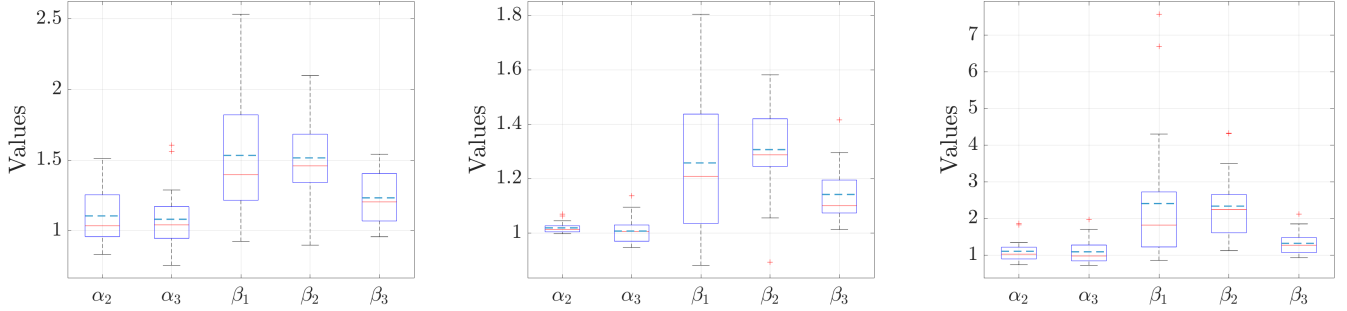


Fig. 3: Boxplot for α_i and β_i as introduced in Theorem 5. The setup of the boxplot is the same as Figure 2. **Left:** Gaussian generation; **Middle:** Hadamard generation; **Right:** Uniform generation.

independently with equal probability, i.e., 50% chance for -1 and 50% chance for 1.

- **Uniform generation:** Set $\mathcal{T} = \mathcal{T}_1 \bullet \mathcal{T}_2 \bullet \mathcal{T}_3 \bullet \mathcal{T}_4$, where each entry of $\mathcal{T}_i \in [0, 1]^{r_{i-1} \times 100 \times r_i}$ is sampled independently from a uniform distribution over the interval $[0, 1]$.

For every tensor $\mathcal{T}$ generated using the above methods, we compute the left and right singular matrices $\mathbf{W}_{\mathcal{T}_{(i)}}$ and $\mathbf{V}_{\mathcal{T}_{(i)}}$ for each unfolding. The index sets $I_i \subseteq I_{i-1} \otimes [100]$ with $I_0 = \{1\}$ and $J_i \subseteq [100^{4-i}]$ are sampled uniformly without replacement. The values of $\alpha_{i,t}$, α_i and β_i are then calculated according to (5) and (7).

For each problem setup, we repeat the experiment 20 times. The results are reported as boxplots in Figure 2, where the legend is detailed in the figure caption. Similarly, Figure 3 reports the boxplots for α_i and β_i .

As shown in Figures 2 and 3, the values of $\alpha_{i,t}$, α_i , and β_i are relatively small with high confidence across all experiments. This empirical observation demonstrates that even with indices generated by simple uniform sampling without replacement, the subtensors well preserve the essential properties of the original tensor, i.e., incoherence and condition number, to some extent for all three random data generation methods.

V. CONCLUSION AND FUTURE WORK

This paper presents a pilot study of property inheritance in subtensors formed by fiber-wise sampling under the tensor train (TT) setting. By focusing on subtensors that preserve the TT rank of the original tensor, we establish theoretical results that elucidate the inheritance of essential tensor properties such as incoherence and condition number. We numerically evaluate the values of key parameters from the theorems and show that the tensor properties are well preserved with simple uniform sampling without replacement. This paper provides a deeper understanding of the relationships between tensors and their subtensors, offering valuable analytic tools for advancing efficient and scalable tensor analysis.

Future work will include a detailed theoretical discussion of the bounds for key parameters with different sampling strategies. Utilizing those properties of subtensors, we aim to further optimize tensor dimensionality reduction methods. Additionally, we plan to extend the results to other tensor decomposition frameworks.

ACKNOWLEDGMENT

This work was partially supported by NSF DMS 2304489. The authors contributed equally.

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